Finite-Time Convergence of Competitive Graph Recoloring via Discrete Lyapunov Functions

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Abstract

We study an asynchronous competitive recoloring process on a finite graph, in which each active vertex selfishly changes color only when doing so strictly reduces the number of same-colored neighbors. Although globally minimizing monochromatic edges is equivalent to a MAX-k-CUT objective and is NP-hard in general, the local dynamics admit a simple exact potential. We prove that every accepted recoloring decreases the number of monochromatic edges by exactly the local improvement, and therefore the process reaches a pure Nash equilibrium in at most $|E|$ accepted recolorings. Simulations on sparse Erdős–Rényi and Barabási–Albert graphs support the deterministic bound and show substantially faster average-case convergence.

Keywords: graph coloring, best-response dynamics, potential game, discrete Lyapunov function, MAX-k-CUT

1. Introduction

Graph recoloring under local incentives appears in distributed coordination, interference avoidance, and networked resource assignment [6, 7]. In such settings, a natural objective is to reduce monochromatic edges, but the corresponding global optimization problem is computationally difficult: minimizing monochromatic edges is equivalent to maximizing bichromatic edges, i.e., to the MAX-k-CUT objective, which is NP-hard in general [4, 5]. This

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motivates the analysis of distributed local heuristics whose behavior can be certified without solving the global problem.

The game-theoretic view is equally natural. Each vertex is a player, each color is an action, and each player seeks to reduce its local conflict cost. Pure Nash equilibria and potential methods are classical tools for such local incentive systems [1, 2, 3]. The specific rule studied here is a strict best-response recoloring dynamics. The main observation is that the number of monochromatic edges is an exact discrete Lyapunov function, in the spirit of potential-based convergence arguments for networked dynamics [8, 9]. Consequently, selfish local updates cannot cycle and must reach a locally stable coloring after at most $|E|$ accepted moves.

2. Model

Let $G = (V, E)$ be a finite undirected graph, with $n = |V|$ vertices and $m = |E|$ edges. Let $C = \{1, 2, \ldots, k\}$ be a finite color set. A configuration is denoted by

$$c = (c_i)_{i \in V} \in C^V,$$

where c_i is the color assigned to vertex i. For $i \in V$, let $N(i) = \{j \in V : \{i, j\} \in E\}$ be its neighborhood. The local conflict cost of vertex i is

$$d_i(c_i, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{c_i = c_j\},$$

where c_{-i} denotes the colors of all vertices except i. Equivalently, if the colors of the other vertices are fixed, then for any $\gamma \in C$,

$$d_i(\gamma, c_{-i}) = \sum_{j \in N(i)} \mathbf{1}\{\gamma = c_j\}.$$

The dynamics are asynchronous. At each activation, a single vertex i is selected. If there exists a color that strictly reduces its local conflict cost, vertex i changes to a best-response color

$$c_i^+ \in \arg\min_{\gamma \in C} d_i(\gamma, c_{-i}),$$

and the update is accepted only when

$$d_i(c_i^+, c_{-i}) < d_i(c_i, c_{-i}).$$

Otherwise the configuration is unchanged. In what follows, a *step* means one accepted recoloring.

Define the global conflict potential

$$\Phi(c) = \sum_{\{u,v\}\in E} \mathbf{1}\{c_u = c_v\} = \frac{1}{2}\sum_{i\in V} d_i(c_i, c_{-i}).$$

Thus $\Phi(c)$ is exactly the number of monochromatic edges.

3. Finite convergence

Lemma 1 (Exact potential drop)**.** *Let c and c^+ be two configurations that differ only at one vertex i. Suppose vertex i changes from color $\alpha = c_i$ to color $\beta = c_i^+$ and strictly reduces its local conflict cost, while $c_{-i}^+ = c_{-i}$. Then*

$$\Phi(c^+) - \Phi(c) = d_i(\beta, c_{-i}) - d_i(\alpha, c_{-i}) \le -1.$$

Proof. Only edges incident to i can change their conflict status, since all other vertices retain their colors. Hence every edge not in $\{\{i,j\} : j \in N(i)\}$ contributes the same amount to $\Phi(c^+)$ and $\Phi(c)$.

Before the update, the number of conflicting edges incident to i is

$$d_i(\alpha, c_{-i}) = \sum_{j\in N(i)} \mathbf{1}\{\alpha = c_j\}.$$

After the update, the number of conflicting edges incident to i is

$$d_i(\beta, c_{-i}) = \sum_{j\in N(i)} \mathbf{1}\{\beta = c_j\}.$$

Therefore

$$\Phi(c^+) - \Phi(c) = d_i(\beta, c_{-i}) - d_i(\alpha, c_{-i}).$$

The strict-improvement rule gives $d_i(\beta, c_{-i}) < d_i(\alpha, c_{-i})$. Since both quantities are integers, their difference is at most -1. ■

Theorem 1 (Finite convergence)**.** *For any finite graph $G = (V, E)$, any finite color set C, and any initial configuration $c(0) \in C^V$, the competitive recoloring dynamics terminate after at most $|E|$ accepted recoloring steps. More precisely, if T denotes the number of accepted recolorings before termination, then*

$$T \le \Phi(c(0)) \le |E|.$$

The terminal configuration is a pure Nash equilibrium, equivalently a fixed point of the strict best-response dynamics.

Proof. By Lemma 1, every accepted recoloring decreases Φ by at least one. The potential is integer-valued and nonnegative because it counts monochromatic edges:

$$0 \leq \Phi(c) \leq |E| \qquad \text{for every } c \in C^V.$$

Thus a sequence of accepted recolorings can have length no larger than the initial potential value. Consequently,

$$T \leq \Phi(c(0)) \leq |E|.$$

No infinite sequence of accepted recolorings is possible.

At termination, no vertex has a color in C that strictly lowers its local conflict cost. Hence no individual vertex can improve by a unilateral recoloring while the other vertices remain fixed. This is precisely a pure Nash equilibrium for the local conflict-cost game and a fixed point of the stated dynamics. ■

4. Empirical validation

Theorem 1 gives a deterministic worst-case upper bound. To illustrate its practical scale, we simulated randomized asynchronous updates on Erdős–Rényi and Barabási–Albert graphs [10, 11] with $k = 3$ colors and constant average degree 8. For each family and each $n \in \{100, 200, 400, 800\}$, we ran 1000 independent trials, giving 8000 runs in total.

No run failed to converge, and no run violated the bound $T \leq \Phi(c(0)) \leq |E|$. The largest observed value of $T/|E|$ was 0.1779, while the average across graph sizes and families was approximately 0.121. Since these graph families are sparse, $|E| = \mathcal{O}(|V|)$; the observed convergence is therefore essentially linear in the network size for these experiments.

Remark 1. *The observed speedup is consistent with the randomized asynchronous schedule: activation orders that approach the deterministic worst case are rarely sampled in standard sparse graph ensembles. The experiments suggest behavior closer to $\mathcal{O}(|V| \log |V|)$, and nearly linear in the tested range, but the formal guarantee used here remains the deterministic edge bound in Theorem 1.*

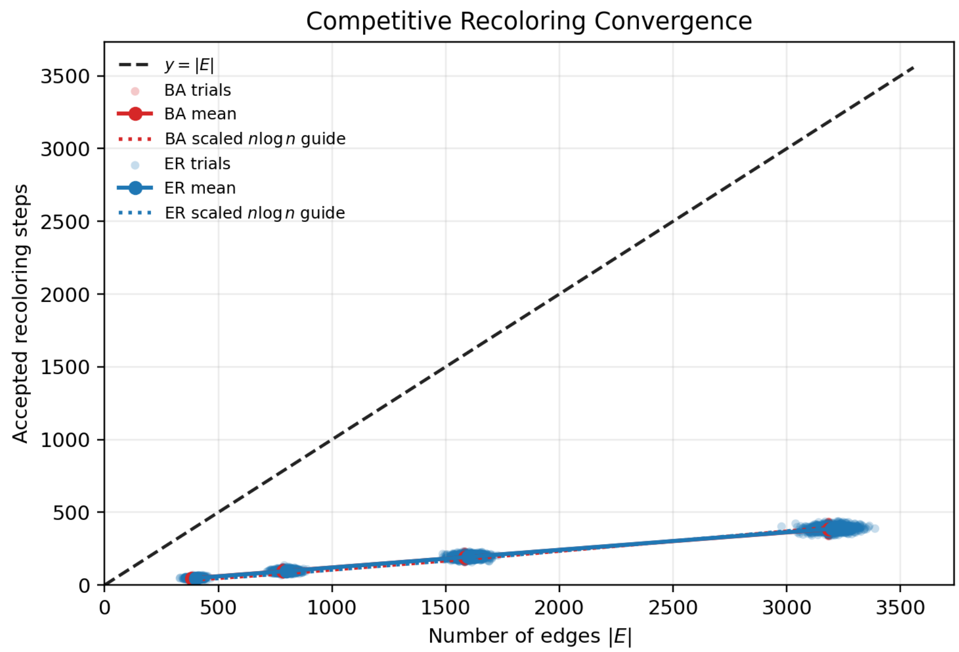

Figure 1: Accepted recoloring steps versus the deterministic edge bound. Each point is one trial on a sparse Erdős–Rényi or Barabási–Albert graph. The dashed line is $T = |E|$. The empirical trajectories remain far below the worst-case Lyapunov bound.

5. Conclusion

Competitive graph recoloring admits an exact discrete Lyapunov function: the number of monochromatic edges. This yields a concise finite-time convergence proof and a deterministic accepted-move bound of $T \leq \Phi(c(0)) \leq |E|$. The simulations confirm the bound and indicate much faster typical convergence on standard sparse random graph families.

The result is intentionally local: it does not require a central coordinator, global optimization oracle, or assumptions on the activation order beyond single-vertex asynchronous updates. Since global minimization of conflicts is intractable in general, the value of the theorem is to certify that a purely selfish rule still has a finite, monotone path to equilibrium. This separates the achievable distributed guarantee from the harder question of finding a globally optimal coloring, and it explains why the empirical behavior can be studied cleanly through the same potential.

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